



RITA LEVI-MONTALCINI

Italian neurophysiologist
1909–2012

Rita Levi-Montalcini's training as a doctor left her full of wonder about the beauty and strangeness of life. The question that puzzled her most of all was how a single cell, the size of a speck of dust, grows into a complex animal made up of billions of different cells!

While in college, Rita learned how a cell can divide to form two new cells, which go on to divide themselves in a process called mitosis. No one could tell her though how these cells became different from one another, each doing different jobs. How did a cell *know* it was supposed to be a nerve cell, for example, rather than a blood cell or a skin cell? It became Rita's mission to find out. She began by studying collections of cells, or tissues, from tiny, unhatched chicken embryos still growing in their eggshells.

The chicken and the egg

Rita studied chicken embryos as they grew from a single, tiny cell into a baby chick. She tried to figure out what caused certain cells to develop into nerve cells.

Rita noticed that if she removed certain tissues from a chicken embryo, the nerves didn't grow in the normal way.